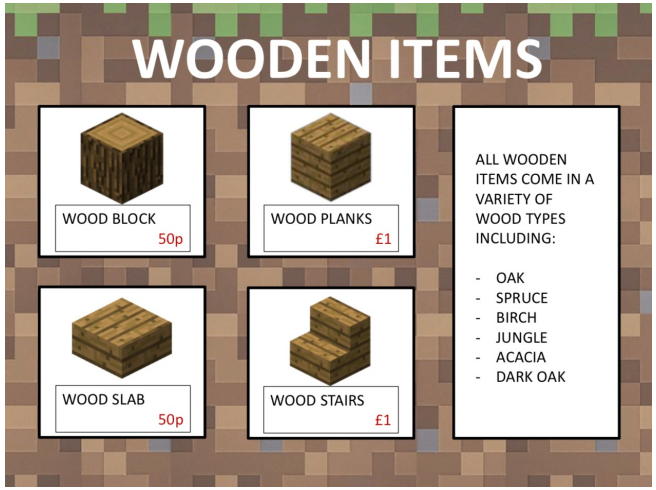
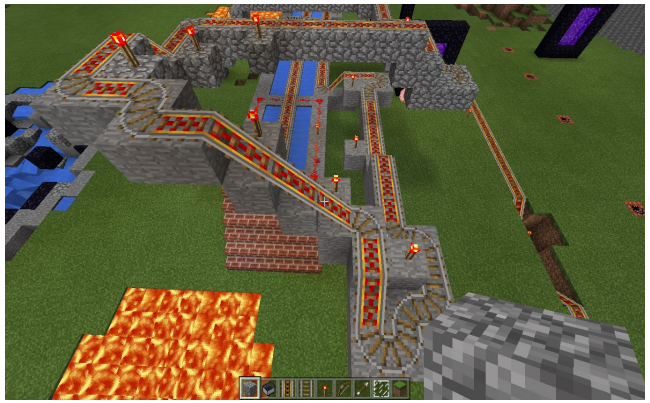
Mathematics, Computer Science, and Education

NUI Galway Taster Day


WOODEN ITEMS
WOOD BLOCK
50p
WOOD PLANKS
£1
WOOD SLAB
50p
WOOD STAIRS
£1
ALL WOODEN ITEMS COME IN A VARIETY OF WOOD TYPES INCLUDING:
- OAK
- SPRUCE
- BIRCH
- JUNGLE
- ACACIA
- DARK OAK



THE MAP OF MATHEMATICS

PURE MATHEMATICS

FOUNDATIONS

FUNDAMENTAL RULES

MATHEMATICAL LOGIC

$p \Rightarrow q$

CONSISTENT SET OF AXIOMS?

NOPE

GÖDEL INCOMPLETENESS THEOREMS

THEORY OF COMPUTATION

0 0 0 1 1 B 0

q_1

P≠NP?

COMPLEXITY THEORY

SET THEORY

A ∩ B

CATEGORY THEORY

$X \xrightarrow{f} Y$, $Y \xrightarrow{g} Z$, $g \circ f$

NUMBER SYSTEMS

CARDINAL NUMBERS

$\aleph_0$ ALEPH NULL

OCTONION

$\{e_0, e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$

QUATERNION

$a+bi+cj+dk$

PI

π

EXPONENTIAL

e

COMPLEX NUMBERS

3, i, 4+3i, -4i

PRIME NUMBERS

3, 11, 47, 907

INFINITY

∞

REAL NUMBERS

$-4\pi, \sqrt{2}, e$

RATIONAL NUMBERS

$-7, \frac{1}{2}, 2.32$

INTEGERS

...-2, -1, 0, 1, 2...

NATURAL NUMBERS

1, 2, 3, 4, 5...

ARITHMETIC

$+ \; - \; \times \; \div$

NUMBER THEORY

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

COMBINATORICS

PARTITION THEORY

TREE

GRAPH THEORY

GROUP THEORY

PERMUTATION GROUP

ORDER THEORY

FACTORS OF 30

MEASURE THEORY

STRUCTURES

LINEAR ALGEBRA

MATRICES

$\begin{pmatrix} 6 & 7 \\ -3 & 2i \end{pmatrix}$

ALGEBRA

$x^2-4x-8=5x+28$

$x^2-9x-36=0$

$(x+3)(x-12)=0$

EQUATION

$y=mx+c$

VECTORS

$\vec{x}$

SPACES

TOPOLOGY

MÖBIUS STRIP

DIFFERENTIAL GEOMETRY

1 HOLE (GENUS 1)

GEOMETRY

CHANGES

COMPLEX ANALYSIS

FRACTAL GEOMETRY

MANDELBROT SET

DYNAMICAL SYSTEMS

BUTTERFLY EFFECT

CHAOS THEORY

FLUID FLOW

ECOSYSTEMS

$\sin(\theta)$ $\cos(\theta)$

PYTHAGORAS

TRIGONOMETRY

VECTOR CALCULUS

CALCULUS

DIFFERENTIAL

GRADIENT $= \frac{dy}{dx}$

INTEGRAL

DIFFERENTIAL EQUATIONS

AREA $= \int_2^9 f(x)\,dx$

ORIGINS

PERSIA c.820

ALGEBRA

c.1730 MATHEMATICAL NOTATION

INDIA c.628

FIRST ZERO 0

$e^{i\pi}=-1$

COUNTING

NEGATIVE NUMBERS

$-8 \; \pi$

50,000 BCE

EGYPT FIRST EQUATION 3000 BCE

CHINA 200 BCE

GREECE 600-300 BCE

APPLIED MATHEMATICS

CRYPTOGRAPHY

ENCRYPT

DECRYPT

PUBLIC KEY

PRIVATE KEY

COMPUTER SCIENCE

TURING MACHINE

MACHINE LEARNING

CAT

```
while awake:
    do_science()
    if self.tired():
        awake=False
self.repair_brain()
```

OPTIMIZATION

PROBABILITY

BAYES' RULE

$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

STATISTICS

MATHEMATICAL FINANCE

GAME THEORY

ECONOMICS

NUMERICAL ANALYSIS

ENGINEERING

CONTROLLER

SYSTEM

SENSOR

CONTROL THEORY

MATHEMATICAL PHYSICS

THEORETICAL PHYSICS

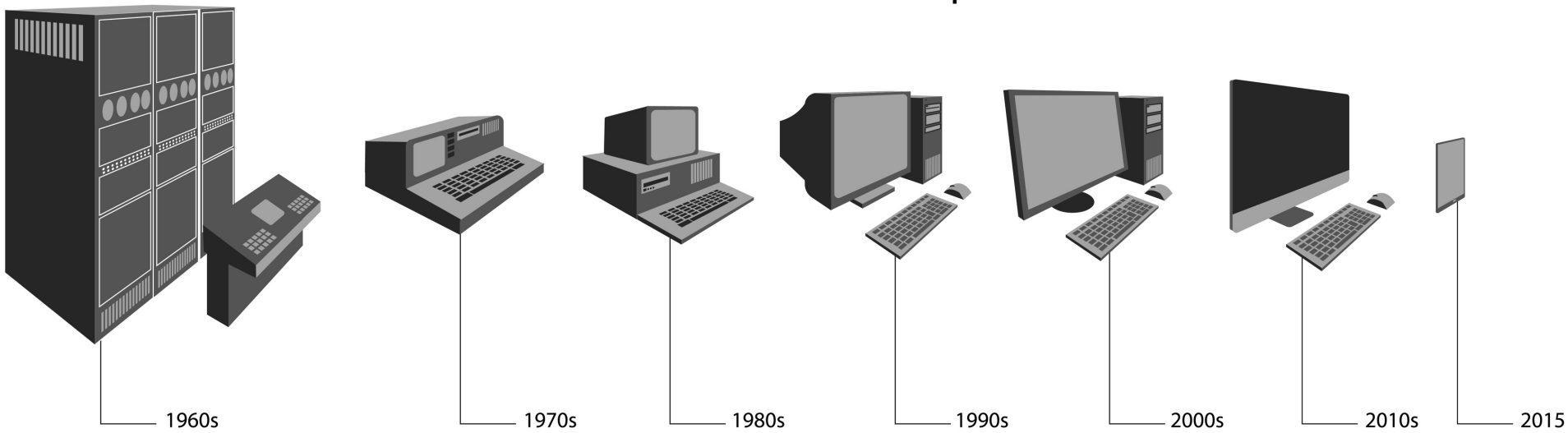
MATHEMATICAL CHEMISTRY

BIOMATHEMATICS

BY DOMINIC WALLIMAN © 2017

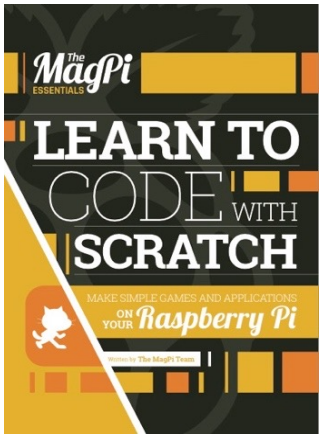
YOUTUBE: THE MAP OF MATHEMATICS

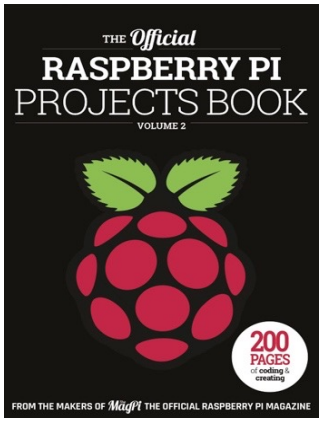
Evolution of computers

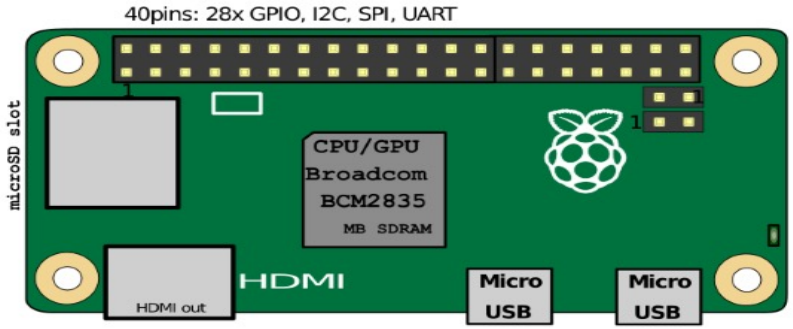


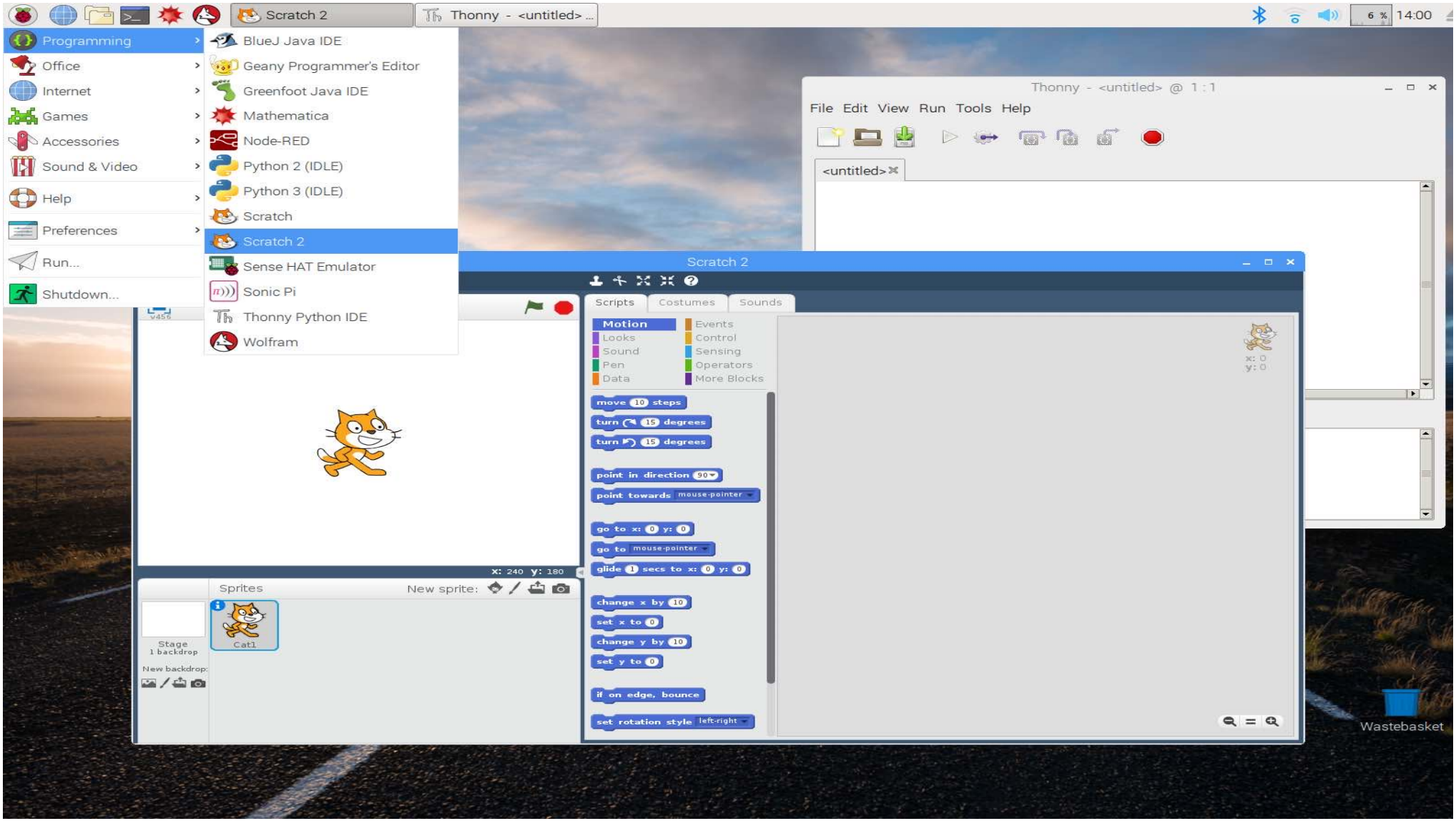
- https://www.youtube.com/watch?v=QvyTEx1wyOY

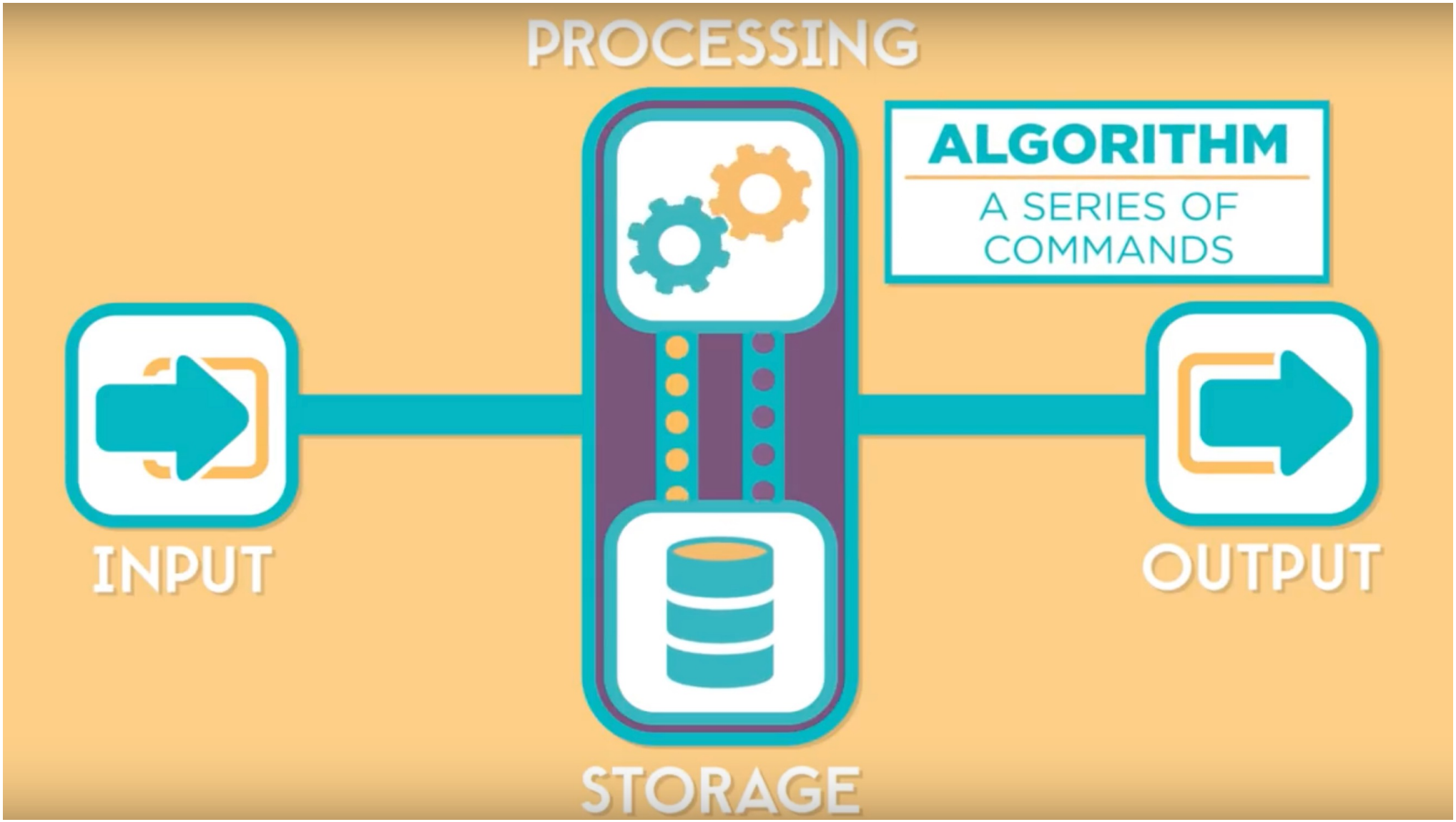
Raspberry Pi®








Scratch 2
Thonny - <untitled> ...
14:00
Programming
Office
Internet
Games
Accessories
Sound & Video
Help
Preferences
Run...
Shutdown...
BlueJ Java IDE
Geany Programmer's Editor
Greenfoot Java IDE
Mathematica
Node-RED
Python 2 (IDLE)
Python 3 (IDLE)
Scratch
Scratch 2
Sense HAT Emulator
Sonic Pi
Thonny Python IDE
Wolfram
Thonny - <untitled> @ 1 : 1
File Edit View Run Tools Help
<untitled>
Scratch 2
Scripts
Costumes
Sounds
Motion
Looks
Sound
Pen
Data
Events
Control
Sensing
Operators
More Blocks
move 10 steps
turn 15 degrees
turn 15 degrees
point in direction 90
point towards mouse-pointer
go to x: 0 y: 0
go to mouse-pointer
glide 1 secs to x: 0 y: 0
change x by 10
set x to 0
change y by 10
set y to 0
if on edge, bounce
set rotation style left-right
x: 240 y: 180
Sprites
New sprite:
Stage
1 backdrop
New backdrop:
Cat1
Wastebasket


PROCESSING
ALGORITHM
A SERIES OF COMMANDS
INPUT
OUTPUT
STORAGE

New Words!

Algorithm

Say it with me: Al-go-ri-thm

A list of steps that you can follow to finish a task

Program

Say it with me: Pro-gram

An algorithm that has been coded into something that can be run by a machine

Programmer \\ Hacker \\ Software Engineer \\ Coder \\ Computer Scientist \\ Developer

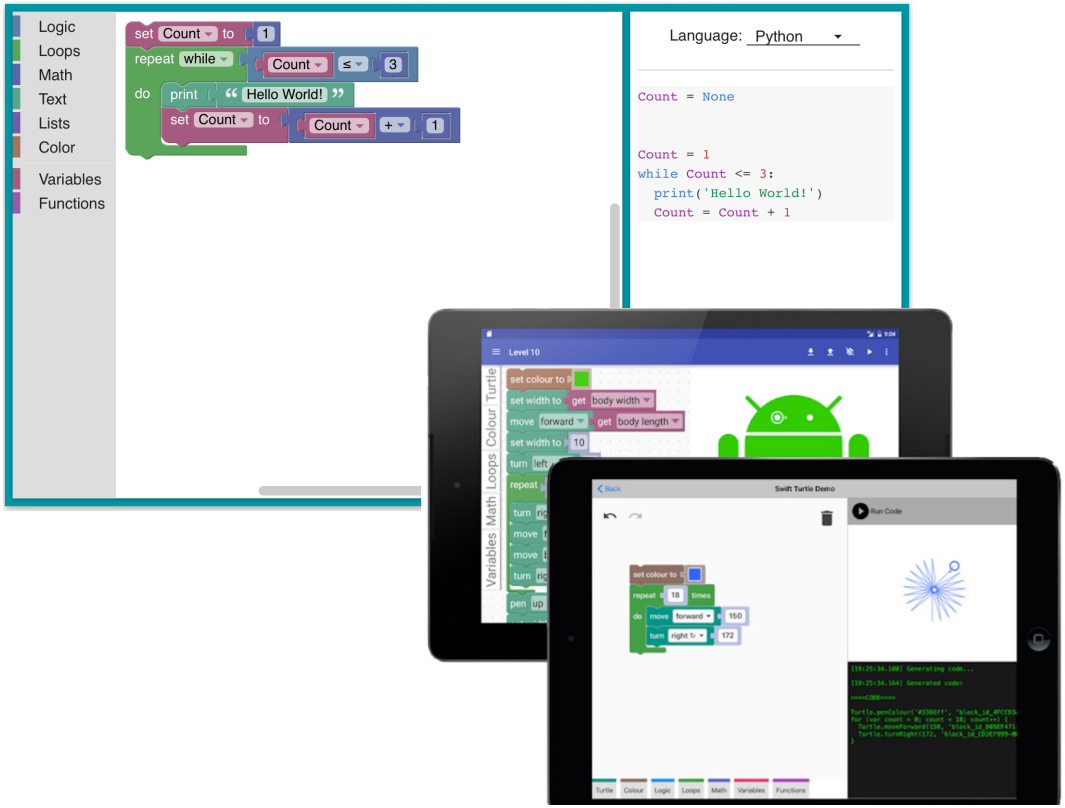
The Math Process

1. Math teaches understanding and communication through abstract language.
2. Math teaches how to work with algorithms.
3. Math teaches students how to analyze their work.

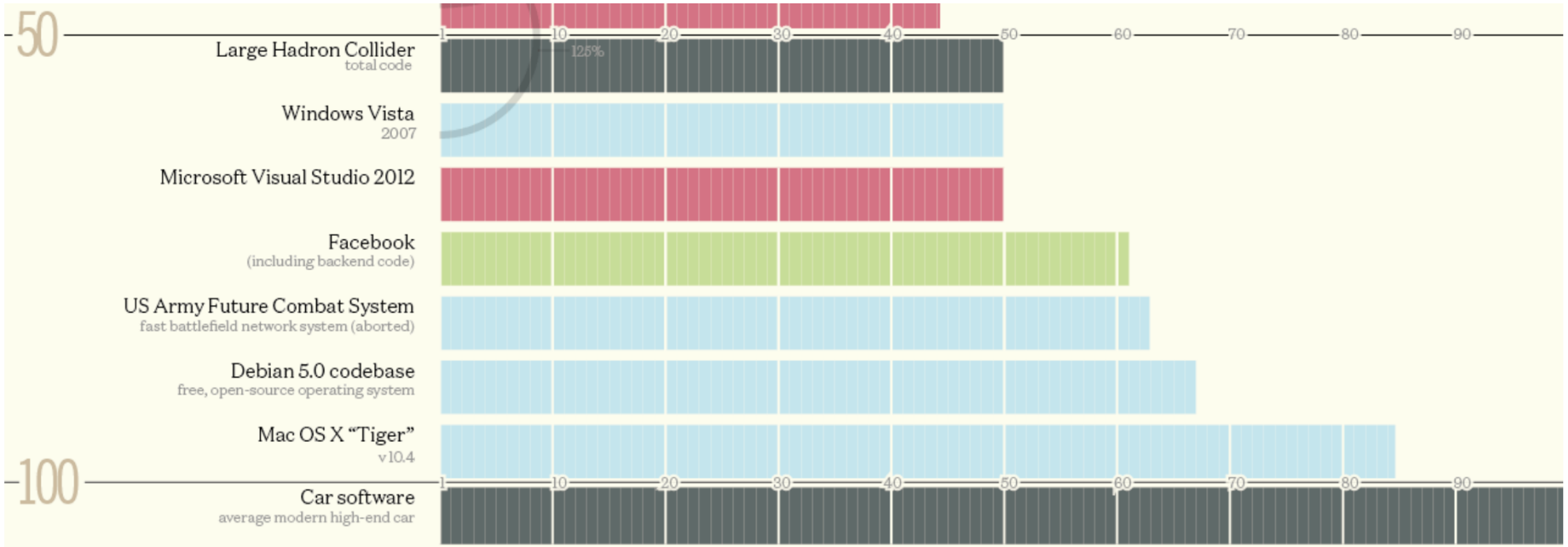
SCRATCH

Blockly Google

A Visual Programming Language



How Many Millions of Lines of Code Does It Take?



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Learn the fundamentals of Computer Science with drag & drop blocks. Create your own drawings and games.

Express Course

An introduction to computer science: combines the best of our elementary school curriculum for older students.

Ages: 9-18

Learn JavaScript, HTML, and CSS

App Lab

Start with App Lab: an introductory programming environment where you can design an app, code with blocks or JavaScript to make it work, then share your app in seconds.

Learn more

Game Lab

Ready to go further? Game Lab is a more complex programming environment where you can make animations and games with characters that run, jump, fly and more.

Learn more

Web Lab

Web Lab is a programming environment where you can make simple web pages using HTML and CSS. Design your web pages and share your site in seconds.

Learn more